

Project Design Phase

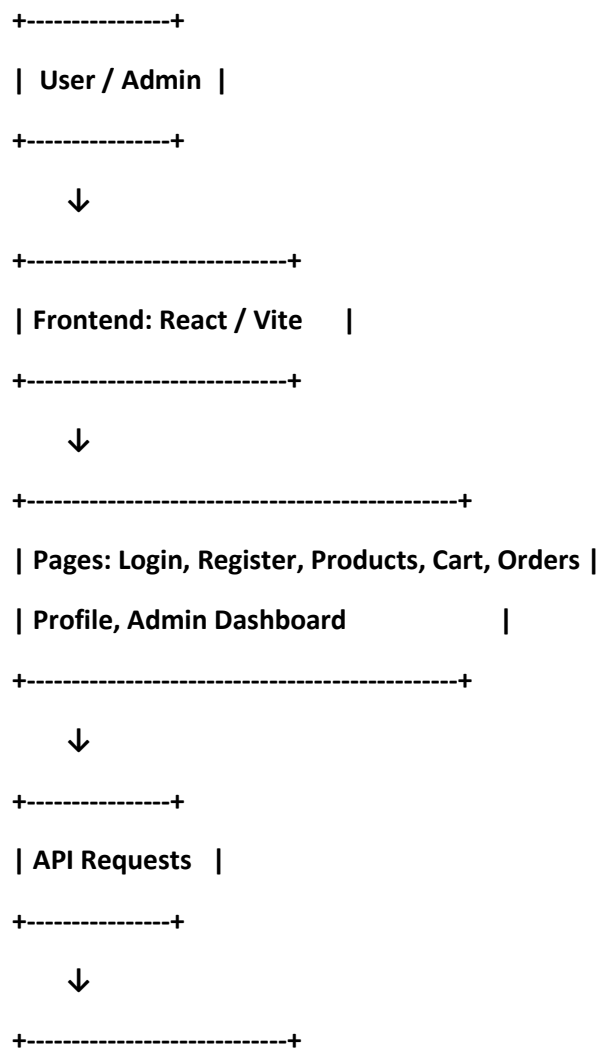
Solution Architecture

Date	18 June 2026
Student Name	Narendula Jashwanth
Project Name	ShopEZ – Full Stack E-Commerce Platform Using MERN Stack
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.



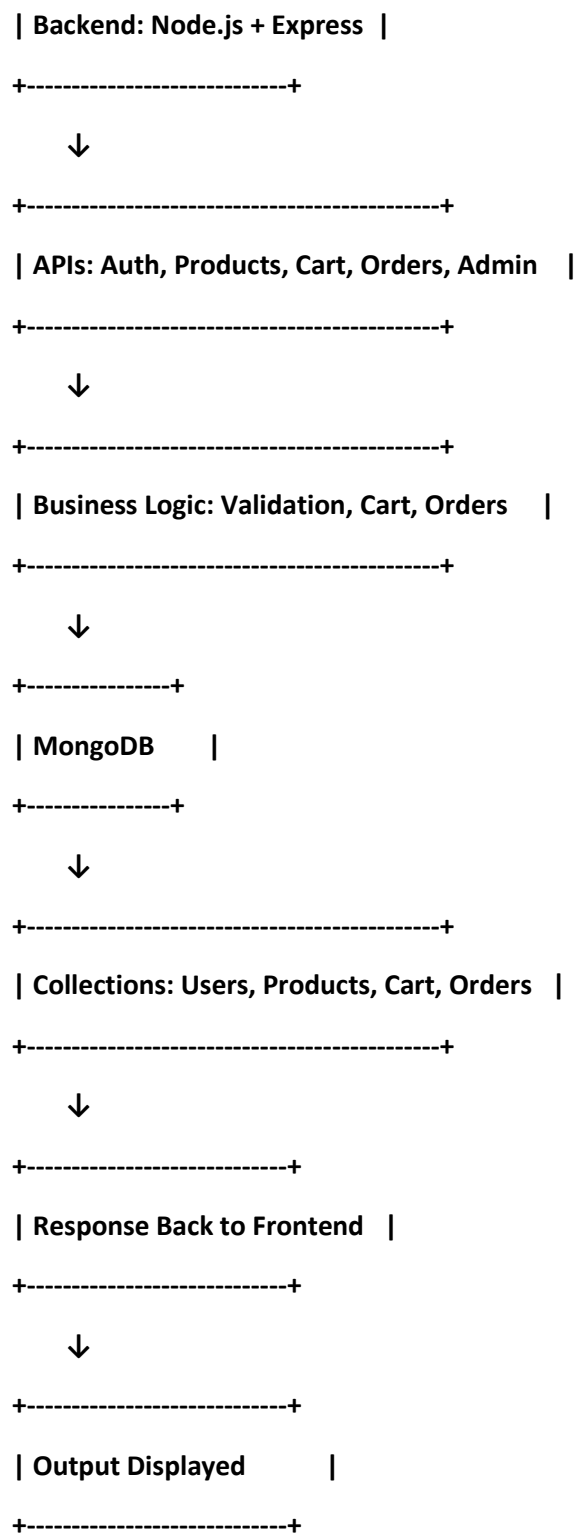


Figure 1: Architecture and data flow of the voice patient diary sample application

System Overview

Field	Description
Project Name	ShopEZ – E-Commerce Web Application

Field	Description
Project Type	Individual Project
Purpose	To provide a simple online shopping platform where users can view products, add items to cart, place orders, and manage their profile.
Users	Online shoppers and admin/shop owner
Main Modules	User Authentication, Product Management, Cart Management, Order Management, User Profile, Admin Dashboard, Database
Frontend	React / Vite application used to display pages and user interface
Backend	Node.js and Express.js server used to handle APIs and business logic
Database	MongoDB database used to store users, products, cart details, and orders
Output	A functional e-commerce web application with user and admin features

Component Description

Component	Description
User / Customer	Registers, logs in, views products, adds products to cart, places orders, and checks order history.
Admin	Logs into the admin dashboard and manages products, users, and orders.
Frontend	Provides the user interface for login, registration, products, cart, orders, profile, and admin dashboard.
Backend	Handles API requests, authentication, product operations, cart operations, and order processing.
Authentication Module	Provides secure login and registration for users and admin.
Product Module	Displays product list and product details to users. Admin can add, update, and delete products.
Cart Module	Allows users to add products, update quantity, remove items, and view cart total.
Order Module	Allows users to place orders and view order history. Admin can manage customer orders.
Profile Module	Allows users to view and update their personal details.
Admin Dashboard	Provides admin control for managing products, users, and orders.

Component	Description
Database	Stores all project data such as users, products, cart items, and orders.
API Layer	Connects the frontend with the backend and transfers data between user interface and server.

Data Flow Explanation

1. User opens the ShopEZ frontend application.
2. User registers or logs in using email and password.
3. Frontend sends the request to backend authentication APIs.
4. Backend validates the user details and connects with MongoDB.
5. User views products from the product page.
6. Product data is fetched from the products collection.
7. User adds products to the cart.
8. Cart data is stored and updated in the cart collection.
9. User places an order.
10. Order details are stored in the orders collection.
11. Admin logs in and manages products and orders through the admin dashboard.